

Lab 1 — Exercise: Notebooks & Markdown

ITCS 3162 — Introduction to Data Mining

Name: *your name here* **Date:** *today's date*

Complete the exercises below. For each task:

- **Code tasks** → fill in the code cell
- **Written answers** → edit the markdown cell directly (look for `YOUR ANSWER:` prompts)

When you're done, run **Kernel → Restart & Run All** to make sure the whole notebook executes top-to-bottom without errors, then download the `.ipynb` and submit via Canvas.

Exercise 1 — Running cells (5 pts)

Run the cell below. Then **modify** it so that it prints your full name and your major.

```
# TODO: change the strings to your name and major
name = "Daniel"
major = "Computer Science"
print(f"Hi, I'm {name} and I'm studying {major}.")
```

Hi, I'm Daniel and I'm studying Computer Science.

Exercise 2 — Variable persistence (10 pts)

Write code in the next cell that creates two variables, `a` and `b`, with any numeric values you like. Then in the cell after that, print their sum, difference, and product on three separate lines.

```
# TODO: define a and b
a = 10
b = 20
```

```
# TODO: print a+b, a-b, a*b on separate lines
print(a+b)
print(a-b)
print(a*b)
```

30
-10
200

Exercise 3 — Markdown formatting (15 pts)

Double-click this cell and replace the placeholder content below with:

1. A level-2 heading with the text "About me"
2. A bulleted list with three items: your favorite class, a hobby, and a goal for this course
3. One sentence containing at least one **bold** and one *italic* word
4. An inline code reference to `pandas`
5. A link to a website you find useful

About Me

- Favorite Class: Backend Application Development
- Hobby: Board Games
- Learning to better analyze and understand data to solve real world problems

I am **very** excited to learn a lot about *data* in this next month, specifically when using tools like `pandas`.

A link: [Quick references for different coding languages](#)

- Usefull for quick reminders for different syntax for a variety of programming languages

Exercise 4 — Math notation (10 pts)

Replace this cell's content with the LaTeX rendering of the **slope-intercept form** of a line, displayed (not inline). Then below it, write one sentence using inline math to explain what m represents.

$$y = mx + b$$

The m in this case represents the slope of the line.

Exercise 5 — A small table (10 pts)

In the cell below, create a markdown table listing **three data mining tasks** we'll cover this semester. Columns: **Task**, **Type** (supervised / unsupervised / association), and **Example use case**.

Task	Type	Example use case
Spam Detection	supervised	Email is flagged as spam if it contains words like FREE MONEY
Stock Prediction	unsupervised	Based in previous trends, predict a certain stock's price at a specific time
Product Recommendation	association	An e-commerce web site gives recommended or often bought together options when a customer adds something to the cart

Exercise 6 — Reflection (10 pts)

In 2–4 sentences, describe one thing you found confusing about notebooks and one thing you find useful compared to writing a regular Python script.

One confusing thing I found, was in the markdown formatting part, referencing pandas, I didn't realize what that meant right away and need to google what that means exactly. Use full thing about notebooks, is that running python code is very simple, gives you the output right away without hassle.

Submission checklist

- ☐ Replaced your name and date at the top
- ☐ All code cells run without errors
- ☐ All **(YOUR ANSWER)** prompts have been replaced
- ☐ Ran **Restart & Run All** before exporting
- ☐ Downloaded as **.ipynb** and uploaded to Canvas